

Jawaharlal Nehru Govt. Engineering College, Sundernagar, Distt. Mandi (H.P)

Program	B.Tech.	Semester	3rd
Subject Code	CSPC-312	Subject Title	Python programming
Mid Semester Test (MST) No.	2	Coordinator	Aditi Mahajan
Max. Marks	50	Time Duration	2 hour
Date of MST	6/11/2024 M	Roll Number	

All Questions are compulsory

Q1 (Ten parts) COs: CO3, 4 Marks: 10*3=30

I. How is list different from tuples and dictionaries?

II. Create empty dict object in two different ways and create a tuple with single element.

III. What will be the output of the following Python code snippet?

```
d1 = {"john":40, "peter":45}
d2 = {"john":466, "peter":45}
d1 > d2
```

What will be the output of the following Python statement?

```
>>> "abcd"[2:]
```

IV. What will be the output of the given three Python code?

```
a = {1:"A", 2:"B", 3:"C"}
for i in a:
    print(i, end=" ")
```

```
a = {1:"A", 2:"B", 3:"C"}
a.items()
```

```
test = {1:'A', 2:'B', 3:'C'}
test = {}
print(len(test))
```

V. Explain split() and join() string object methods. Write output of code: "Welcome to Python".split()

VI. What will be the output of the following Python code snippet?

```
print('The sum of {0} and {2} is {1}'.format(2, 10, 12))
```

If A = {5,6,7}, write value of A after A.add(5) is executed?

VII. Write a short python script to remove duplicate elements from a list.

VIII. Explain read(), readline() and readlines() in python files with help of example.

IX. What will happen if you open binary files in text mode?

X. Explain the difference between add() and update() method w.r.t to set object with help of code.

Q2. Explain the mutability and concept of key-value pairs of dictionaries in Python. How can you add, update or remove key-value pairs from a dictionary?

CO3 10

Q3. How are tuples created in Python. What are the two methods that are used in Tuple? What are some common use cases for using tuples?

CO3 10

Course Outcomes (CO), Students will be able to

- 1. Python Proficiency:** To have a understanding of the basic principles and concepts of Python covering its operators, data types, expressions, type conversions and basic inbuilt functions.
- 2. Control flow statements & Functions:** To learn how to write and use various loops and decision statements in Python. To learn how to write functions and pass arguments in Python.
- 3. Iterables:** Using string, lists, tuples, set and dictionaries and applying various operations and methods to them in Python programs
- 4. Files, CSV files:** To able to read and write different types of files in Python and be able to work with CSV files.